

Learning goals: Basics of shell scripting continued: Introduction to sed and awk

NOTE: Please do all your work in a new folder called `lab13`. Save the script for each exercise separately as `question1.sh`, `question2.sh` and so on. For all questions with functions, please store the output of the function in a variable and then print the value of that variable in the main script. At the end of the lab, please 'push' your scripts to GitHub!

EXERCISES

- (1) Create a text file containing the following text and do the following actions using `sed`.

```
Python is a high-level, interpreted programming language
that is widely used for web development, data science, automation,
artificial intelligence, and more.
It was created by Guido van Rossum in 1991 and
has grown into one of the most popular programming languages in the world.
Key Features of Python
Readable and simple syntax - Python code often looks like plain English,
making it beginner-friendly.
Interpreted language - You don't need to compile Python;
you can run your code directly.
Cross-platform - Works on Windows, macOS, Linux, and many other systems.
Extensive libraries - Python has built-in modules and
third-party packages (via PyPI) for tasks like math,
data analysis, networking, and visualization.
Versatile use cases - From small scripts to large-scale applications.
```

- (i) Print only the lines where the word `and` appears.
 - (ii) Change all instances of the word 'language' into 'lang'.
 - (iii) Delete lines containing the word 'is'.
 - (iv) Find a way to insert line numbers at the beginning of each line
 - (v) Remove 1st 2 lines only
 - (vi) Find a way to print every other line in the above file.
 - (vii) Substitute first instance of 'Python' by 'python' and 'language' into 'lang' using one `sed` command.
- (2) `awk` exercises. Add the following data to a simple text file.

```
Ram 20 Biology
Sita 23 Maths
Gita 26 Physics
Suresh 23 Chemistry
```

The second column is the marks obtained by the student in that particular subject.

- (i) Find a way to print the people's names who are scored less than 25 in their subjects.
 - (ii) Print the student whose Physics score is given.
 - (iii) Find a way to rewrite the data into a csv format in a new file called `data2.csv`
- (3) A common file type you will encounter in bioinformatics is called a FASTA file. For example, few lines of a FASTA file looks like this:

```
>seq1
ATGCTAGCTAGAGGTCTAGG
>seq2
TTAGGCATGCATTAAGTCATTA
```

- (i) Using `awk` remove the lines containing `>`. These are called header lines.

- (ii) Using `sed` convert the DNA sequence to RNA sequence.
- (iii) Using `sed` replace `seq1` (called the sequence ID) by `human_gene`.
- (4) Visit the PDB website, and download the protein with ID `1HK0` in the *pdb* format. With this, use `awk` to extract the atoms of all phenyl alanine residues of this protein and save it in a new file called `PHE_atoms.xyz`